

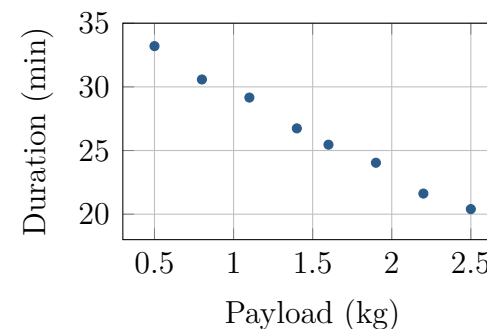
# Two Payloads, One Regression Line

Topic 2.6 · TODAY'S PROBLEM

Suppose eight drone tests with payloads from 0.5 to 2.5 kg produced the shown data and the model  $\hat{y} = 36.0 - 6.4x$ , where  $x$  is payload and  $\hat{y}$  is predicted flight duration in minutes. A proposed mission carries 3.5 kg and requires 12 minutes of flight.

**Ravi:** Approve it: the model predicts more than the required 12 minutes, and the fitted data are close to a line.

**Mina:** Before deciding, compare the support for predictions at 1.8 and 3.5 kg and identify evidence the safety decision still needs.



## FIRST TAKE (5 min, on your sheet)

Would you approve now, delay approval, or approve conditionally? Cite one number or plot feature.

- DOK 1** (a) Identify the explanatory and response variables, slope, intercept, and observed payload range.
- DOK 2** (b) Interpret the slope. Predict duration at 1.8 and 3.5 kg, classify each prediction, and calculate the modeled buffer above 12 minutes.
- DOK 3** (c) Evaluate both approaches using the observed range, the distance from 3.5 kg to that range, the modeled buffer you computed, and the in-range plot. Decide what to do, explain whether the two predictions have equal support, specify evidence needed before approval, and state the consequence of the rejected reasoning.

**Video + follow-along:** [u2\\_lesson6\\_live.html](https://u2_lesson6_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

